

v1.6.0

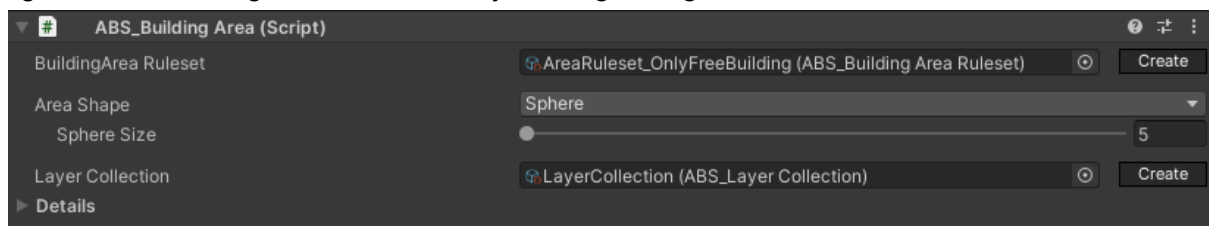
The goal of this documentation is to give a short demo and some information about the new features of the version 1.6.0. This document does not explain everything in detail. For a detailed explanation read the core documentation of the asset! Also note that not every change is included by this document. Mostly the new features and those changes are worth mentioning.

Removed features	2
BuildingArea Settings	2
Raycast Settings	2
CanSnapUnderground property	2
New Features	3
New Position Validations	3
Grounded	3
Build On Top Of Element	3
Others	5
Destroy Only Mode	5
Attachment Connection for FreeBuilding	5
Structure stability feature for the AdvancedGirdBuilding	6
Small changes what worth to mention	7
Gizmos	9
Tools	10
Building Element Creator	10
Scene Helper	10
Objects	10
Analyzer	11

Removed features

BuildingArea Settings

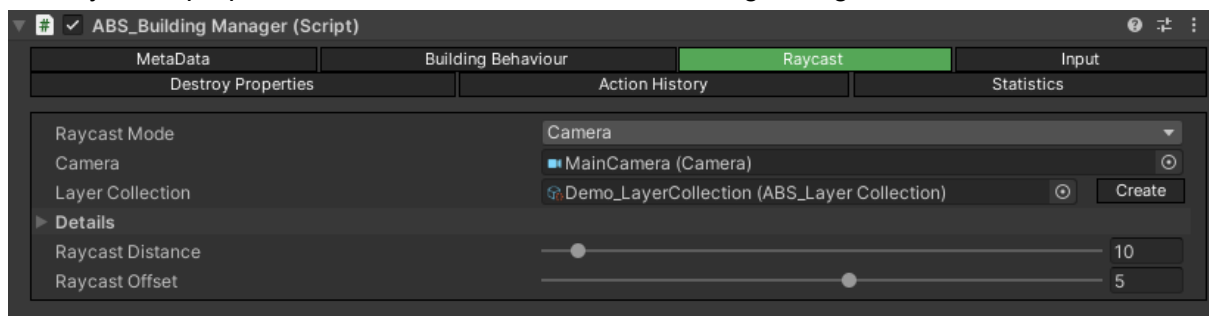
The BuildingAreaSettings has been removed. You as the developer could create a ruleset for the rules and a settings file for the settings and use the settings for the areas. The main idea was that you can use the same ruleset for different settings and also you can use the same settings for different areas. There was no problem with that but I tried to decrease the necessary steps for the setup of the AdvancedBuildingSystem and I tried to reduce the necessary file count. These settings contained only 2-3 properties so it is not a big effort to set it again and again for every BuildingArea. So I decided that I will remove this settings file to reduce the amount of files that should be created and from now the properties can be set right on the Building Area. Functionally nothing changed!



Raycast Settings

Just like the BuildingAreaSettings the Raycast settings has been used in only one place and it only contained just a few properties. So I removed it too to decrease the filecount of the AdvancedBuildingSystem.

The raycast's properties are available from on the BuildingManager:



CanSnapUnderground property

The CanSnapUnderground property has been removed because it seems totally unnecessary because if there is an enabled validation why only one has this basically a negation property which ignores the validation. It seems to me I should add this option for every validation or remove this to be more consistent.

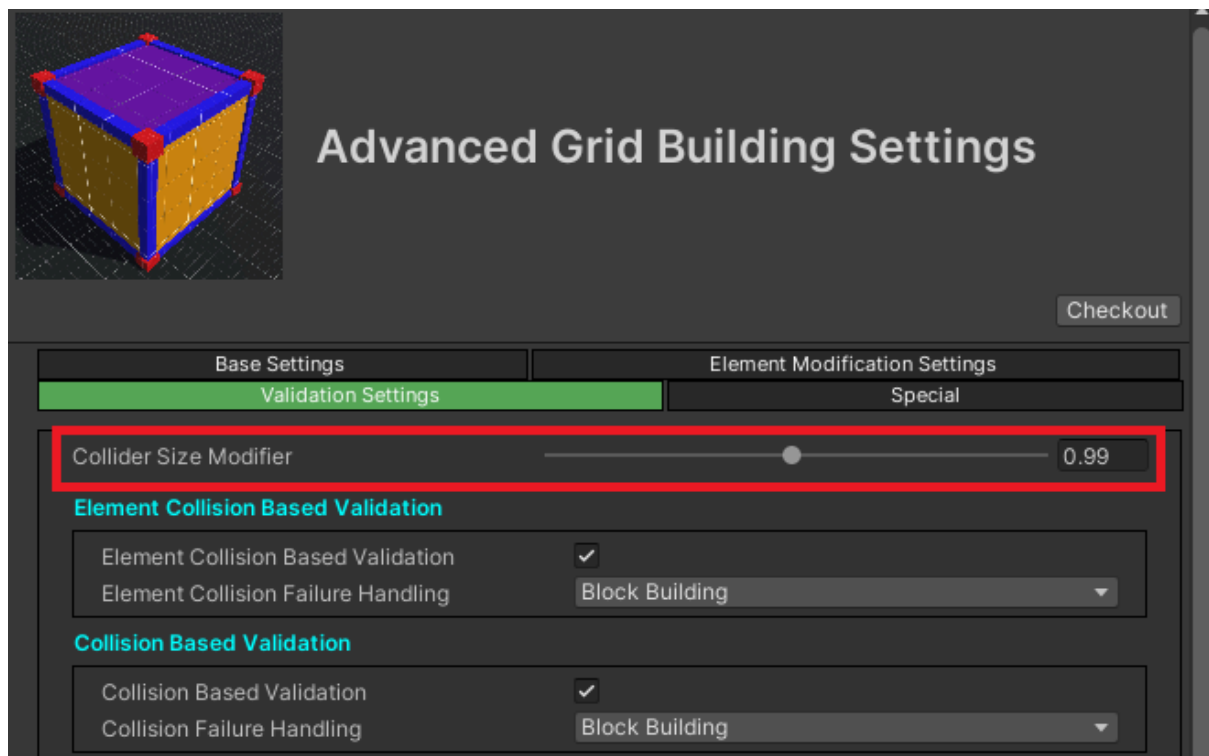
New Features

Position Validations

Collider Size Modifier

From now a multiplier can be set for the validations. If the validation is using a collision check it will be multiplied for this modifier. It is recommended to keep smaller than 1. It can decrease or increase the size of the collider.

The main use case is when an element is aligned to another on the top of it like a piece of furniture inside the house. The collision check can find the elements under it and block its placement.



Grounded

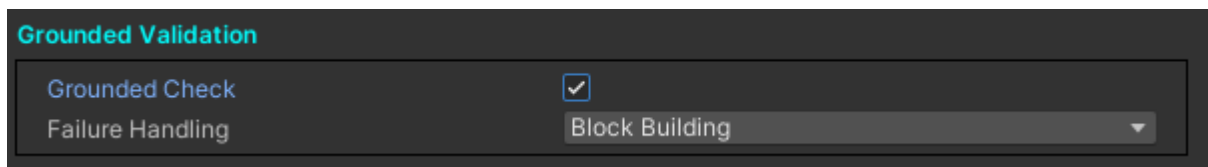
A new ground based validation has been introduced into the asset next to the Underground validation and the Buildable Ground Validation.

This is not supported only by the BasicGrid algorithm because it has its only Ground validation logic.

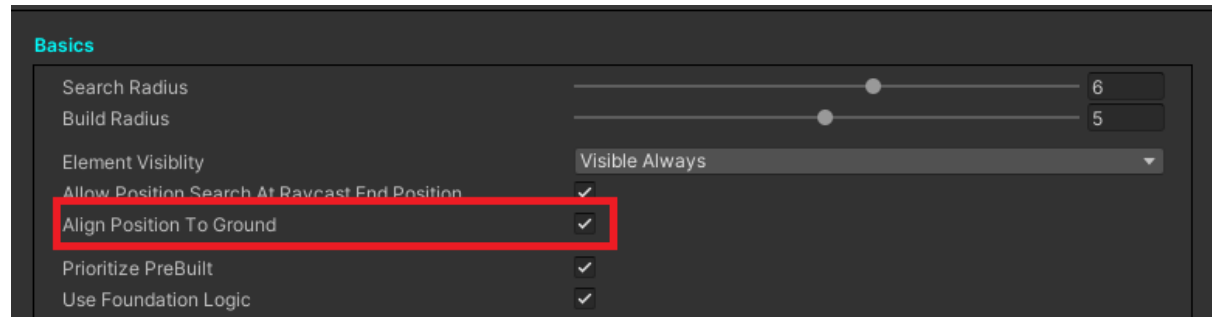
Meanwhile the Underground checks if the element is under the ground with a downwards raycast and the BuildableGround Validation checking with multiple downwards raycasts if the element has more than one grounded position the new Grounded check uses collision check to ensure that the element is grounded.

So the new Grounded Validation performs a collision check and if it collides with the ground in any point of the collider (not just downwards) then the element is grounded.

Note that even if the collider did not collide with the ground but the element is aligned to the

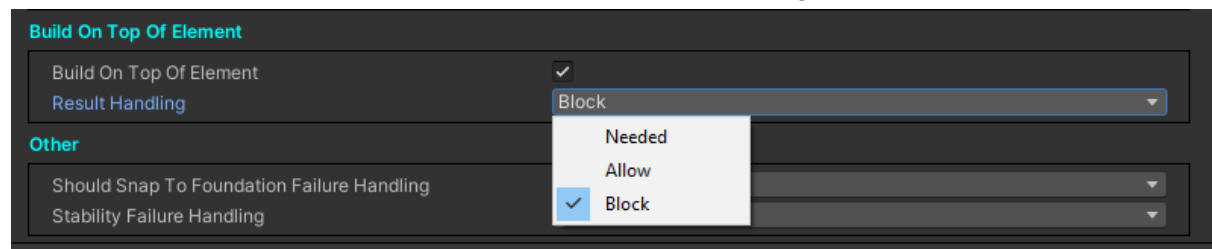


ground by another feature the element counts as grounded.

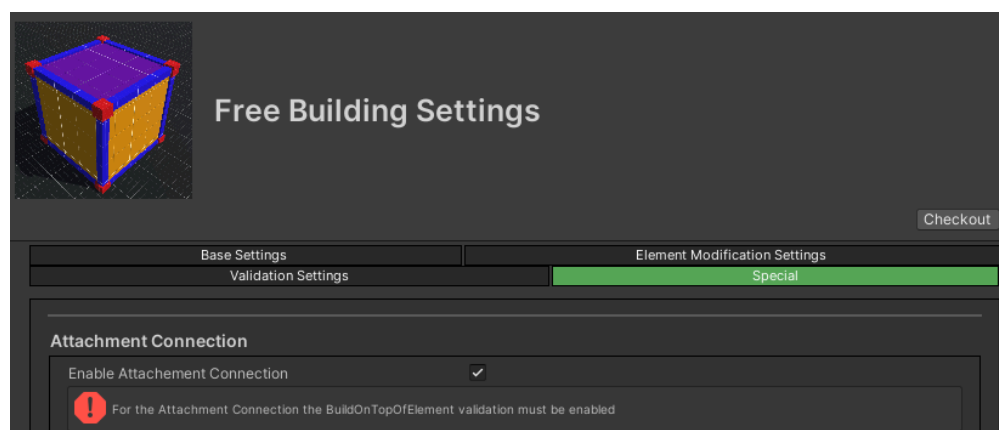


Build On Top Of Element

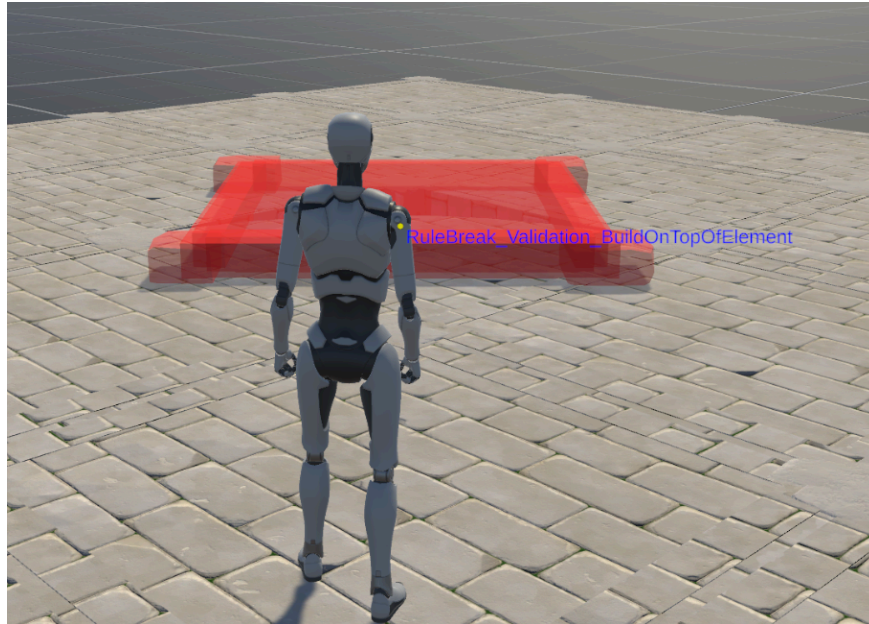
This new validation will perform a raycast downwards like the Building In The Air Validation or the UnderGround Validation and try to check if the object underneath the element is another element. This validation has its unique result handling:



- Needed
 - The element should be on top of another element.
 - The validation fails if the element is not on the top of another element.
 - The main idea is that with this validation setup you can ensure that an element like a furniture can be built on another element onto the building.
- Allow
 - The element can be on the top of another element.
 - The validation can not block the building of the element. Basically this setup can not failed.
 - If it is set the algorithm will check and get the information that the new element's position is on the top of another element. This is a must have information for the new Attachment connection feature which will be explained later. For this setup you will get errors on the UI if something missing:

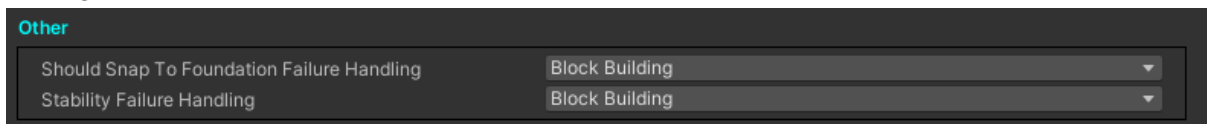


- Block
 - The element can not be on the top of another element.
 - The validation fails if the object underneath the new element is another element.
 - The main idea is that if the snapping of the AdvancedGridBuilding fails the fallback FreeBUilding algorithm can restart the building on the top of the other building. With this block setup you can avoid that:



Others

Even if the following validation features are not setup like the other validations the features can block the building of the element so they got a failure handling here the Validation Settings:



Destroy Only Mode

From now the BuildingManager can be activated as only in destroy mode and activating with an element and changing it to destroy are not necessary anymore.

The BuildingManager has the following modes:

```
public enum ABS_BuildingManagerMode
{
    Unkown,
    SimpleBuild,
    DragBuild,
    SimpleDestroy,
    DragDestory,
    SimpleDestroyWithoutElement,
    DragDestoryWithoutElement
}
```

Attachment Connection for FreeBuilding

From now the AdvanceBuildingSystem has a connection system which means that specific connections can be set up between the elements. The connections are reconnected and disconnected during the history redo, undo features and also saved by the persistence feature which means that the load process reconnects the element.

The first connection introduced in this update is the Attachment connection for the FreeBuilding.

If an element is built with FreeBuilding algorithm as the main algorithm the attachment feature can be enabled and then if the element is built on the top of another element the newly placed FreeBuilding element will be attached to the another element. Even if the element is attached to another element the newly placed element will be still under the GlobalFreeBuilding just the connection is saved and handled between the element.

This attachment connection means that if the element that has the attached element has been destroyed then the attached elements will be destroyed too.

For this feature the Built On Top Of Element validation should be enabled and step up.

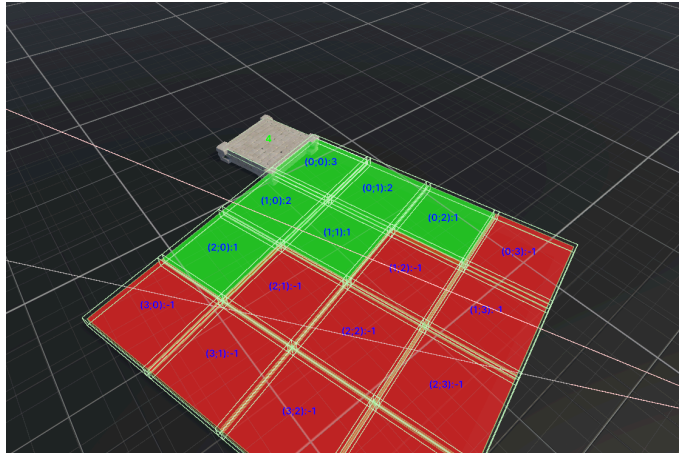
The main idea is that if the player destroys an element like the floor from a building the furniture on that will no longer be flying in the air. They can be destroyed too.

Structure stability feature for the AdvancedGridBuilding

The second connection between the elements is the structure stability connection which is supported by only the AdvancedGridBuilding algorithm. This feature is about the AdvancedGridBuildings calculating its elements' stability which is based on the neighbour element's stability. If an element has been destroyed the stability is recalculated and if another element becomes unstable then the destroy action will destroy those elements too.

The stability works in that way the stable element has maximum stability and every element attached to it calculates its stability from the neighbours and gets the maximum stability as possible.

Here is an example the the stability is level 4 :



Small changes what worth to mention

1. Override change
 - a. In that case when the stability feature is working together with the AdvancedGridBuildingSnapPointRuleset there can be some situation when the placing an element can implicate the destruction of some other element.
 - b. This is the front he rules and the stability but this doesn't look good from the player point of view so from now only those elements can be overridden by others that have the same AdvancedGridSnapPointRuleset.
2. Blocking the input
 - a. The problem is that if the BuildingManager's is inactive and the interface can not be blocked so the Manager can not start with a blocked interface which is a required feature.
 - b. From now the BuildingManger's input interface can be blocked even if the manager is deactivated.
3. ForceFallback
 - a. The problem is that the ForceFallback feature is basically doing the opposite of the foundation feature. So when you use both at the same time and the player tries to build a non-foundation element with forced fallback the 2 algorithm makes the element unable to be placed. In this case the element will be blocked no matter what.
 - b. So from now if the foundation feature is on then only the foundation elements can be built using the ForcedFallback. None of the other elements can be forced to build with freebuilding.
4. Tracker::SetExternalInterface
 - a. The SetExternalInterface function became virtual instead of the abstract so from now it shouldn't be implemented.
 - b. Also it saves the external interface of the BuildingManager so from the code the trackers can reach it with the m_BuildingManager property.

```
////////////////////////////////////  
#region Properties  
////////////////////////////////////  
protected ABS_IBuildingManagerExternalInterface m_BuildingManager = null;  
////////////////////////////////////  
#endregion // Properties  
////////////////////////////////////  
#region ExternalInterface  
////////////////////////////////////  
4 references | Changed locally  
public virtual void SetExternalInterface(in ABS_IBuildingManagerExternalInterface p_ExternalInterface)  
{  
    m_BuildingManager = p_ExternalInterface;  
}  
////////////////////////////////////  
#endregion // ExternalInterface  
////////////////////////////////////
```

5. New ErrorCodes

- a. Because of the new features, especially the new validations, new error codes are introduced.
- b. ABS_SimpleBuildingProcessErrorCode
 - i. RuleBreak_Validation_GroundedCheck
 - ii. RuleBreak_Validation_BuildOnTopOfElement
 - iii. RuleBreak_AdvancedGridBuilding_Stability
 - iv. RuleBreak_FreeBuilding_ShouldAttached
- c. ABS_PersistencyLoadErrorCode
 - i. BuildingElementConnection_MissingBuilding
 - ii. BuildingElementConnection_MissingBuildingElement
- d. ABS_ActionHistoryErrorCodes
 - i. Failed_PositionValidation_NotEnoughSpace
 - ii. Failed_DataValidation_WrongInstanceGuid
 - iii. Failed_DataValidation_WrongPrefabGuid

6. Statistics

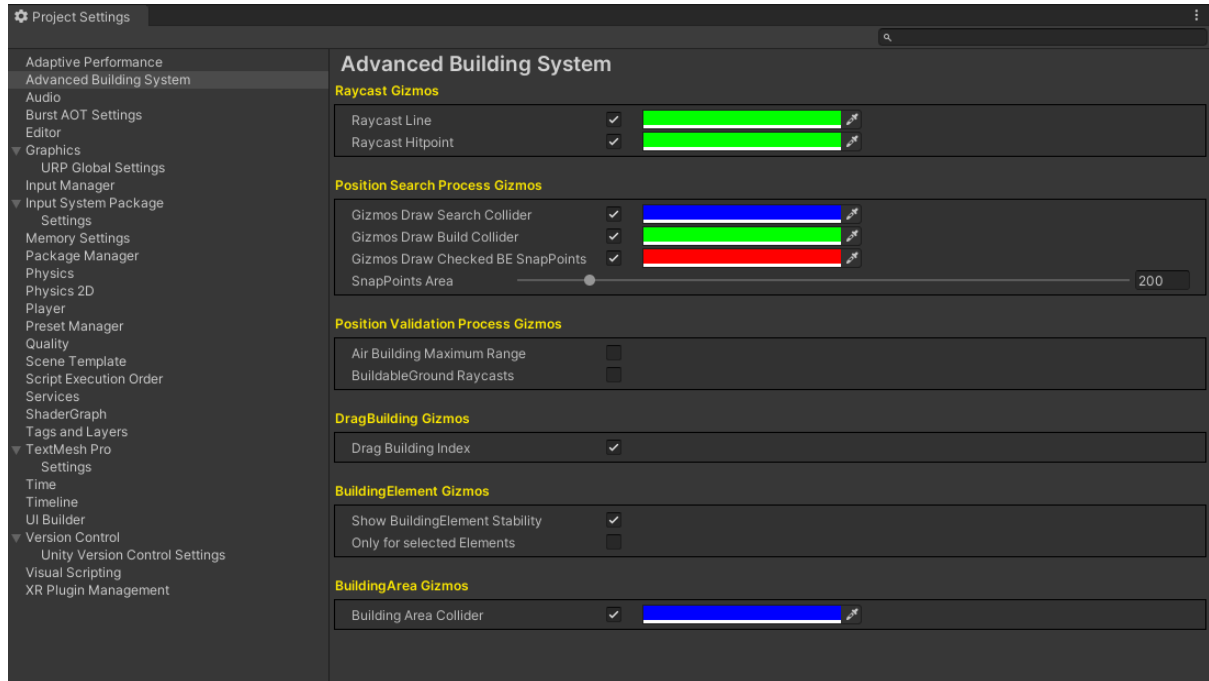
- a. The statistics are changed from the milisecs to nanosecs.

7. BuildingElementList

- a. New checks are introduced

Gizmos

The Gizmos is basically for debugging the asset. It is not needed for everyone so I decided that I will hide them so I removed the Gizmos from the Interfaces of the objects of the AdvancedBuildingSystem. The Gizmos has been centralized and can be found in the Project Settings. Every object of the ABS uses these properties for their Gizmos.



Also some new gizmo has been introduced!

Tools

Building Element Creator

The tool have been aligned with the new features so the copy functionality will copy the following properties additionally to the old ones:

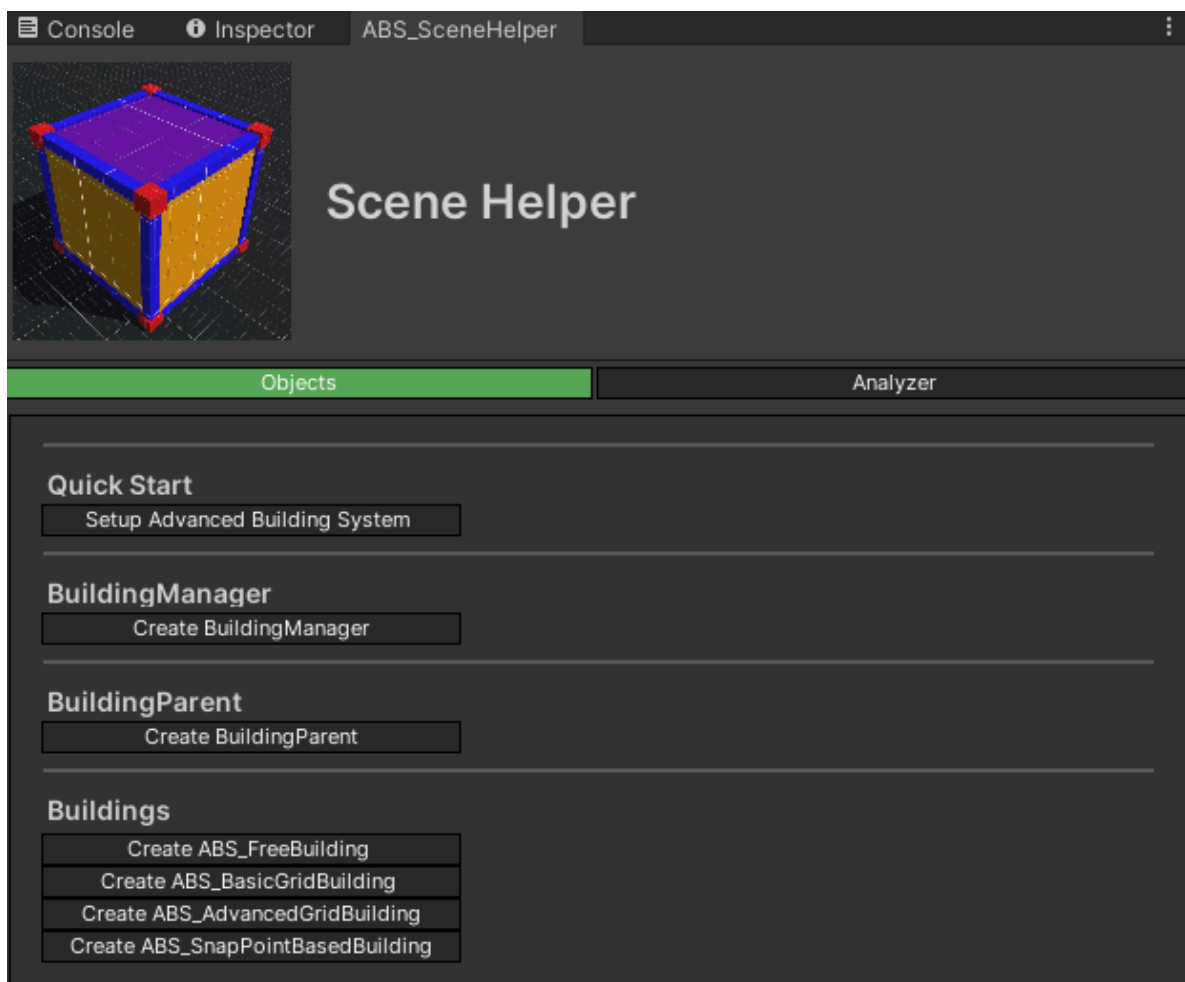
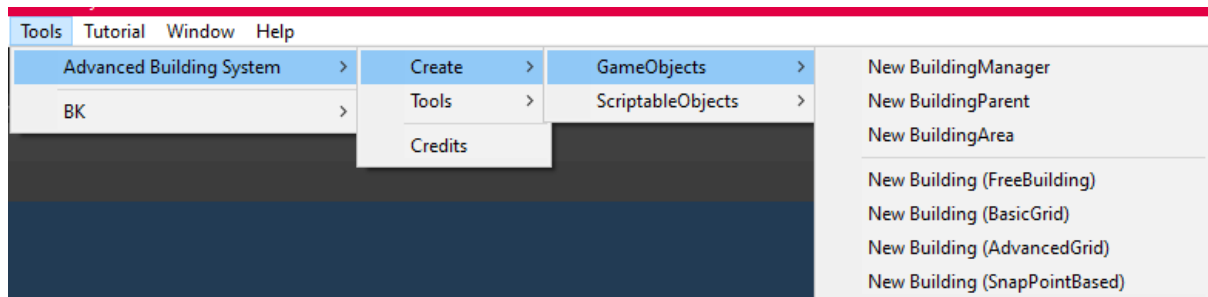
- Can Not Be Attach Target
- Should Attached
- Stable Element
- AdvancedGrid Snap Point Rule Set

Scene Helper

Objects

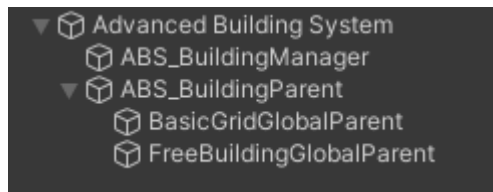
The SceneHelper got a new tab called Objects that can be used to create ABS GameObjects into the scene.

In functionally this is basically the copy of the Create/GameObject section of the Menu :

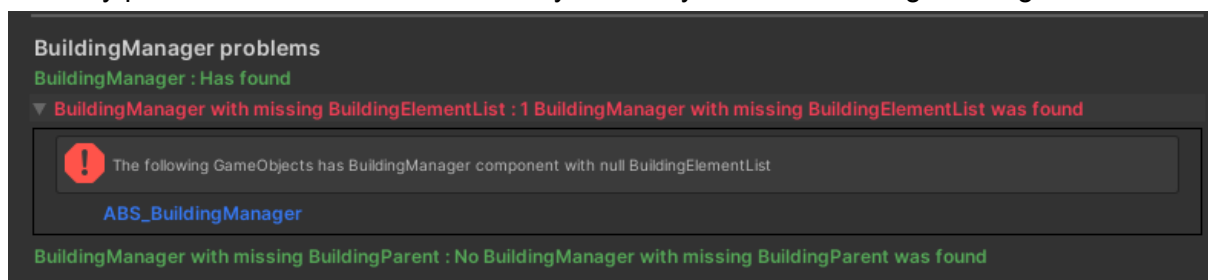


On this tab you as a developer can easily create GameObjects into the scene that is properly set up so the SceneHelper's Analyzer feature should not show any problem with them. Of course it is true only if you use it properly so the recommendations should be followed like the Building's parent object should be a BuildingParent.

Additionally this tool has an extra feature which is the "Quick Start". With this quick start option the SceneHelper creates a whole Advanced Building System setup into the scene.



The only problem what should be shown by the analyzer is the missing BuildingElementList:



Analyzer

The analyzer got some new checks:

- BuildingElement problems
 - AdvancedGrid problems
 - Wrong grid position
 - Wrong grid rotation
 - BasicGrid problems
 - Wrong grid position
- BuildingParent problems
 - No BuildingParent in the scene
 - BuildingParent with missing Global FreeBuilding
 - BuildingParent with missing Global BasicGridBuilding
- BuildingManager problems
 - No BuildingManager in the scene
 - BuildingManager with missing BuildingElementList
 - BuildingManager with missing BuildingParent
- BuildingArea problems
 - BuildingArea without RuleSet
 - Building Area without LayerCollection